The Way Between

What a turn word holds, what its moves cost, and why every lock opens once the chain may leave the plane

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Abstract

A lattice chain is the word of turns that draws it. On the honeycomb every turn is one bit, so an m-bit word draws a chain of m + 2 beads, valid when it does not cross itself, and the operators of the binary ring become the chain's symmetries: the complement C is the mirror image and JC is the same chain read from its other end. In space the chain's shape is its dihedral word $\delta(x)_i = b_{i-1} \oplus b_i$, cis where adjacent turns agree and trans where they differ. That is the difference reader of *One Inverter*: the shape and one face bit return the word, the mirror is a proper half-turn about the first bond, and every move is a set of torsion flips. A one-bit flip changes the two dihedrals at its pivot and swings the whole tail; the shortest local rewrite, 1001 ↔ 0110, changes two dihedrals four bonds apart and is a crankshaft of two beads; a tail or prefix mirror changes one. A move Hamiltonian that respects the mirror splits into an even and an odd sector, and the odd sector carries a π flux on exactly the components that reach their own mirror images; the flux is the half-turn that a closed loop of shapes leaves on the frame. Locality is paid for in connectivity. The span of the move masks fixes the widths at which the mirror is reachable ($m \equiv 0 \bmod 4$ with hexagon moves, 0 or 1 with the last turn added, every class but 3 with the last two, every m with two at each end), a bead beside the end never moves under the one-ended vocabulary, and every finite local vocabulary tested leaves chains it cannot reach, as the Madras–Sokal theorem on local moves leads one to expect. With three turns free at each end the graph is one piece through m = 21 (826,564 chains); from m = 22 it leaves isolated chains, and after a two-hexagon exchange 4, 4, 10 and 28 locks remain at m = 23 to 26. All 46 open once the chain may leave the plane. A four-bead end gate is two torsions; along the synchronous path the end block turns rigidly about its pivot at angular speed $\sqrt{(2 + \sigma)}$, and with that exact constant every one of the 68 endpoint-valid gates on the locks is certified collision-free with clearance at least 0.977 bond lengths, where the in-plane pivot must bend a bond angle straight and collides on every lock at m = 24 and 26. The locks were a missing degree of freedom, not a missing rewrite. Every statement is checked by one sealed notebook: 44 checks, exhaustive where the text says so.

How the paper runs

The paper is written from its end, as the series has been. Section 1 states the finished object: a 26-turn chain that no planar move frees, and the twist that frees it. Section 2 records why an address cannot serve as a move, which is where the work of 23 October began. Sections 3 to 8 follow the chain from the word that draws it to

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the dimension that opens it; each opens on the hole the previous result left and names what must fill it by its shape and its neighbours. Section 9 states what the construction compiles and what it does not, against the prior art. Results first reached in the GPT and Gemini runs and the integrated manuscript of 23 October, and in the audits that checked them [4, §§ 144–154], are marked with their source, and every one was rerun here with independent code. The statements were hashed before the notebook that checks them first ran (Appendix B); the notebook's own ledger closes the paper (Appendix C). Words are written most significant bit first, and string position 0 is turn 0.

1. The held end

The finished object is one chain, the 26-turn word

$$x^* = 01011110101101011101011110,$$

drawn on the honeycomb from bead 0 at the origin with the first bond pointing east (Figure 1B). It is a ring of hexagons with both four-bead end blocks tucked inside, and in the plane it cannot move. No hexagon rewrite and no flip of its first three or last three turns leaves a self-avoiding chain, no two-hexagon exchange applies, and the two four-bead end moves, the flips of turns 3 and 22, sweep their blocks through the chain on every planar arc. It is one of 46 such locks at widths 23 to 26, and every finite local vocabulary tested leaves locks at some width (Sections 6 and 7).

Out of the plane it opens. The flip of turn 3 changes two dihedrals, about bonds 3 and 4. Turned together in opposite senses, the head block rotates rigidly about bead 4 and lands on $x^* \oplus e(3)$ with every pair of non-incident bonds at least 0.98 bond lengths apart at every instant, a certified bound. The flip of turn 22 opens a second way, two half-turns of the tail in series, each collision-free. The way between two planar states runs through the third dimension.

Four statements carry the paper.

- **The route.** A word of turns is a chain. The ring's operators are the chain's symmetries, the chain's shape is the word's adjacent differences, and the one bit those differences miss is the frame (Sections 3 and 4).
- **The sectors.** A move Hamiltonian that respects the mirror splits in two, and the odd half carries a π flux on exactly the components whose chains reach their own mirror images. The flux is a half-turn of the frame (Section 5).
- **The price.** Local moves pay for locality in connectivity: the span of their masks, a bead that never moves, and the theorem that no finite local vocabulary is ergodic for the self-avoiding walk. Every vocabulary tested here freezes chains at some width (Sections 6 and 7).
- **The way out.** Every local move is a torsion motion in space. The hexagon rewrite is a crankshaft, an end gate is one rigid rotation, and every lock found opens in three dimensions with a certified clearance (Section 8).

Table 1 collects the readings of x^*. Each is printed by the notebook of Appendix C.

Reading	Value	Where
x^*, 26 turns, 28 beads	01011110101101011101011110	Section 1
$C(x^*)$, its mirror image	10100001010010100010100001	Theorem 3.1
dihedral word, bonds 1–25 (1 = trans)	11100011110111100111110001	Theorem 4.1

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Reading	Value	Where
face of x^*; face of $C(x^*)$	0; 1	Theorem 4.2
both3 moves; two-hexagon exchanges	none; none	Theorem 7.5
gate 3: serial corners	both self-intersecting	Proposition 8.4
gate 3: in-plane arcs, 120° and 240°	both collide	Proposition 8.6
gate 3: synchronous, counter-rotating	sampled 1.000; certified 0.984885	Theorem 8.5
gate 3: synchronous, co-rotating	sampled 0.912897; certified 0.886717	Theorem 8.5
gate 22: serial orders	2	Proposition 8.4
gate 22: in-plane arcs, 120° and 240°	both collide	Proposition 8.6
gate 22: synchronous, co-rotating	sampled 1.000; certified 0.977328	Theorem 8.5
gate 22: synchronous, counter-rotating	sampled 0.654657; certified 0.641567	Theorem 8.5

Table 1. The running example. Clearances are minimum distances between non-incident bonds, in bond lengths; a certified value holds at every instant of the path. The smallest certified gate value over all 68 gates on the 46 locks (0.977328) and both smallest sampled values (0.912897 co-rotating, 0.654657 counter-rotating) occur on x^* and its images under C and J.

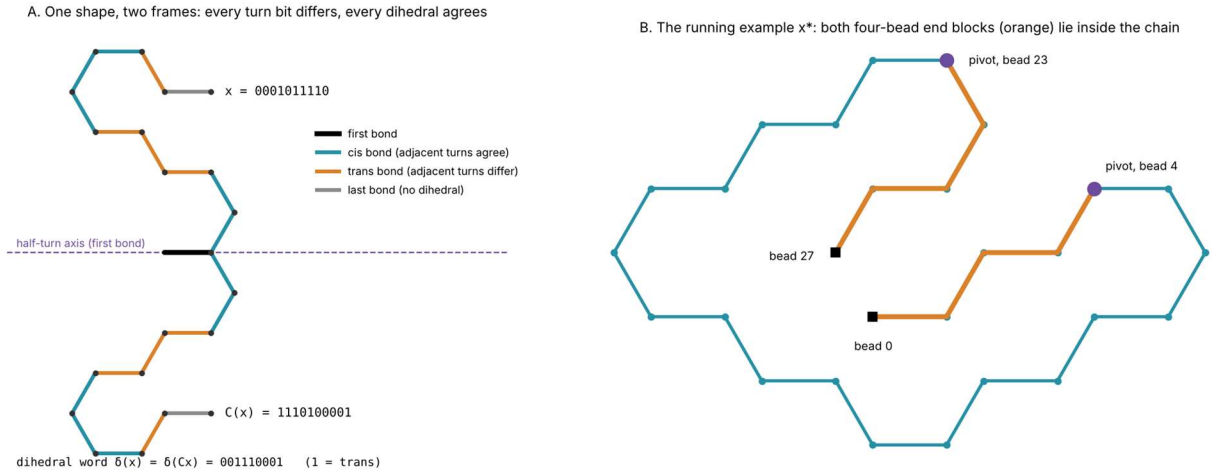


Figure 1. (A) One shape in two frames. x = 0001011110 and its mirror image C(x) differ in every turn bit, and every interior bond is cis (teal) or trans (orange) in both alike; in space the two chains differ by a half-turn about the line of the first bond (dashed). (B) The running example x^*, a ring of hexagons with both four-bead end blocks (orange) inside it. The end gates pivot about beads 4 and 23.

2. Address is not a move

The hole. The ring of all binary words locates every word without enumeration [1], and *One Inverter* gave it a loop and a reader [3]. GPT's run of 23 October asked whether the same machinery folds a chain. It took six-residue rings on the square lattice as the states, the ring order as their addresses, hopping along the ring as the kinetic term, and a penalty on addresses that are not rings. Before any Hamiltonian is written, what must a move be?

Proposition 2.1 (six rings, no move). On the square lattice there are exactly six closed six-step self-avoiding walks with the first bond east, and all six are 2 × 1 dominoes. With the sequence PHPPHP their contact energies

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are −1, −1, 0, 0, 0, 0. Two of them differ at three sites (6 pairs) or at four (9 pairs), never at one, so no single-residue move joins any two. **[proved; RUNS; sealed]**

Proof. A closed walk of six unit steps bounds two unit squares side by side, and five residues of such a domino fix the sixth. ■

So any dynamics among the six rings has to leave them, and the run measures what leaving costs.

Proposition 2.2 (the toy is the ring and its successor). GPT's address order is the ring R_3 = (0, 1, 2, 6, 3, 5, 4, 7) [1, 3], and its kinetic term $K = -\gamma(T + T^{-1})$ hops along the ring successor T, which joins words at Hamming distances 1, 2, 1, 2, 2, 1, 2, 3. **[RUNS; sealed]** T is the address successor, not an adjacency [3, Proposition 6.3]: consecutive catalogue entries differ at three or four residues.

Proposition 2.3 (leakage, two paths, bridges). Take $\gamma = 0.7$, valid addresses 0, 1, 2, 4, 5, 6, padding addresses 3 and 7, and the uniform valid state ψ_0. The early leakage is $\|QK\psi_0\|^2 = 4\gamma^2/3 = 49/75$, and $\|QKP\| = \sqrt{2}\cdot\gamma$. The valid graph is two paths, 0–1–2 and 4–5–6, so with the kinetic term projected onto the valid sector the contact probability never exceeds ½; at t = 1 it is 0.463631173441. The second-order bridges KQK through the padding are $\gamma^2 = 0.49$ on the diagonal at 0, 2, 4, 6 and on the pairs (0, 6) and (2, 4). **[proved; RUNS; sealed]** (GPT, § 144)

Proposition 2.4 (a penalty does not remove the leak). At t = 1 the leakage is 0.324402465, 0.006480472, 0.000151516 and 0.000023742 for padding penalties λ = 0, 10, 50 and 100, and exactly 0 with the projected mixer. At λ = 10, ||[V, K]|| = 10.406248, ||[V, [V, K]]|| = 109.882164 and ||[K, [K, V]]|| = 24.111397, and second-order Strang splitting with r = 4, 8, 16, 32, 64, 128 steps has operator error 0.082770330, 0.018193978, 0.004416348, 0.001096147, 0.000273545, 0.000068356, close to a factor 4 per doubling. On the four-state ring HPPH the contact probability at t = 1 is 0.7717491215. **[RUNS; sealed]** (GPT, §§ 144 and 147)

Proposition 2.5 (a merge leaves a record). A unitary that sends six orthogonal states to one output keeps them orthogonal in its environment: for the SWAP on $\mathbb{C}^6 \otimes \mathbb{C}^6$ with the output fixed, the six environment records are orthonormal. **[proved; RUNS; sealed]** A funnel that forgets which state it came from is not unitary. The way home has to be written somewhere.

Proposition 2.6 (the loop is a twisted translation). As a $2^n \times 2^n$ permutation matrix the loop τ of *One Inverter* is $U_\tau = X_{last}\cdot R$, with R the cyclic shift of the qubits, and $U_\tau^n = X^{\otimes n}$ is the global spin flip. Its spectrum is one full set of L-th roots of unity for each τ-orbit of length L. **[proved; RUNS n = 2–8; sealed]**

τ is the Z_2-twisted translation of a spin ring. Its fermionic twin is the twisted translation of the Majorana chain [11, eq. 3.27], the ring's centre is a domain-wall (ZZ) readout, and the odd parity of *One Inverter*'s Theorem 5.4 is the odd number of domain walls that a twisted ring must carry.

The toy leaves an exact hole. An address is not a move: a move must change a chain into a chain, its inverse must be the same kind of change, and no state on the way may fail to be a chain. What fills the hole is an encoding in which the word is the chain.

3. The input is the route

The hole. The states must be chains, the moves must be chain moves, and the ring's operators must act on both. The encoding is fixed by that shape.

Draw a chain on the honeycomb from bead 0 at the origin, with the first bond pointing east. At each later bead the chain turns left or right by 60°, so the word of turns $b_0 \ldots b_{m-1}$ (0 for left, 1 for right) draws a chain of m + 1

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bonds and m + 2 beads, and every such chain has exactly one word. The input is the route, literally: the word is the line it draws. Write V_m for the words whose chains are self-avoiding, e(j) for the flip of turn j, and use integer coordinates (a, b) in the basis of the bonds at 0° and 60°.

Theorem 3.1 (the encoding). (a) C, the complement, draws the mirror image of the chain across the line of the first bond, and JC draws the same chain read from its other end. (b) The flip e(i) rotates beads i + 2 to m + 1 rigidly by 120° about bead i + 1: a pivot move. (c) $|V_m|$ = 2, 4, 8, 16, 30, 58, 112, 216, 406, 776, 1,472, 2,796, 5,260 and 9,964 for m = 1 to 14. **[proved; RUNS: every word to m = 14, pivots to m = 11; sealed]**

Proof. (a) A reflection reverses the sense of every turn; reading backward reverses their order and their sense. (b) Changing the turn at bead i + 1 from +60° to −60°, or back, changes the heading of every later bond by the same 120°. ■

The ring's four operators are the chain's four symmetries: the identity, the mirror image (C), the reversal (JC) and the mirror image reversed (J). The one-bit edges of the word cube are pivot moves.

Proposition 3.2 (the pivot graph). The graph on V_m whose edges are one-bit flips between valid words is connected for m = 2 to 14, with 4, 12, 32, 70, 160, 356, 776, 1,594, 3,352, 6,890, 14,124, 28,312 and 57,236 edges. **[RUNS; sealed; connectivity at every width OPEN]** (GPT, § 145)

Every edge is its own inverse, and nothing passes through an invalid word: there is no padding and no leakage. In route coordinates this is the pivot algorithm of Lal [5], which Madras and Sokal proved ergodic for the self-avoiding walk [7]; whether their proof covers the honeycomb with this move set is not checked here.

Proposition 3.3 (the mirror is the symmetry of the energy). Score each pair of non-bonded H beads on adjacent sites as −1. C keeps every energy for every sequence, while J and P reverse the order of the residues and need not. On HHPHPPHHPPHP at m = 10 (776 chains, 32 of them with a contact) J and P each change 48 energies and C changes none. **[proved; RUNS; sealed]** (GPT, § 145)

C is therefore the operator that can remain a symmetry of a molecular Hamiltonian with no further assumption.

4. Two readings of one word

The hole. A chain drawn in the plane can be lifted into space without changing a bond length or a bond angle. What does the word say about the lifted chain, and what does it say only about the plane?

Give every bond length 1 and every bond angle 120°. For i = 1 to m − 1 the dihedral about bond i, which joins beads i and i + 1, is the angle between the planes of beads (i − 1, i, i + 1) and (i, i + 1, i + 2).

Theorem 4.1 (the shape is the dihedral word). For a planar chain the dihedral about bond i is 0 (cis) when $b_{i-1} = b_i$ and π (trans) when they differ:

$$\delta(x)_i = b_{i-1} \oplus b_i, \quad i = 1, \ldots, m-1.$$

δ is linear over GF(2) with kernel $\{0, 1^m\}$, so every dihedral word has exactly two framed words, x and C(x). Two valid chains are congruent exactly when their dihedral words agree, and C sends each site (a, b) to (a + b, −b), the proper half-turn about the line of the first bond. **[proved; RUNS: dihedrals of all 32,764 words, m = 2–14; exact distance matrices of all 5,894 valid chains, m = 2–12; sealed]** (half-turn: GPT, § 152; dihedral word: § 154)

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Proof. Consecutive turns in the same sense put beads i − 1 and i + 2 on the same side of bond i, and opposite senses put them on opposite sides. Bond lengths, bond angles and dihedrals fix a chain up to a rigid motion, and the reflection (a, b) ↦ (a + b, −b) of the plane is, in space, the half-turn about its axis. ■

The two readings of a word are shape and frame. The shape is the word's m − 1 adjacent differences; the frame is which of x and C(x) is drawn, and in space the two differ by a half-turn (Figure 1A).

Theorem 4.2 (the reader). Let face(x) = [x > P(x)], with P the partner of the ring [2, 3]. Then face(Cx) = 1 − face(x), so the dihedral word and the face bit together return x. For even m the centre bit of *One Inverter*, the XOR of the two central bits, is the dihedral at the middle bond. **[proved; RUNS: every word, n = 2–16; sealed]**

Proof. P commutes with C, and C reverses the order of the integers, so x > P(x) exactly when Cx < P(Cx). ■

This is *One Inverter*'s reader, n − 1 adjacent differences and one face bit [3, Section 5], met again in space. The shape quotient that § 152 found is not new structure; it is the difference reader, re-addressed.

Proposition 4.3 (moves are torsion flips). $\delta(x \oplus M) = \delta(x) \oplus \delta(M)$, so each move flips a fixed set of dihedrals. The turn flip e(j) flips dihedrals j and j + 1. A hexagon window at i, turns i to i + 3, flips dihedrals i and i + 4. The tail mirror D_k, turns k to m − 1, flips dihedral k alone, and so does the prefix mirror Q_{k-1}, turns 0 to k − 1. In particular $e(j) = D_j \oplus D_{j+1} = Q_{j-1} \oplus Q_j$. **[proved; RUNS m = 6–64; sealed]** (the identity: Gemini, § 151)

A pivot changes two torsions at its vertex and swings everything beyond them; the hexagon rewrite changes two torsions four bonds apart and nothing else; a mirror changes one.

5. Mirror sectors and the frame holonomy

The hole. Let $H = E - \gamma A$ on V_m, with E the contact energy and A the adjacency of a move graph closed under C. C commutes with both (Proposition 3.3). What does the symmetry split, and what does the split carry?

Theorem 5.1 (two sectors). Let U_C permute V_m by C. Then $[H, U_C] = 0$, and in the basis $(|a\rangle \pm |Ca\rangle)/\sqrt{2}$, one vector of each sign for each mirror pair, $H = H_+ \oplus H_-$ with

$$H_\pm = E - \gamma(A_0 \pm A_1),$$

where A_0 counts the edges between representatives a and b and A_1 the edges from a to Cb. The quotient must keep its self-loops (an edge from a to Ca) and its parallel edges (edges to both b and Cb). **[proved; RUNS: pivot graph m = 6, 8; hexagon and hexagon + terminal m = 4, 5, 6, 8; residual below 10^{-12}; sealed]** (GPT, §§ 145–146) At m = 4 the hexagon edge 1001–0110 is a self-loop of the quotient; at m = 5 under hexagon + terminal moves, and at m = 8 under both vocabularies, the quotient carries parallel same-sheet and cross-sheet edges.

Theorem 5.2 (an odd loop exactly where the mirror is reached). Sign each quotient edge +1 if it comes from A_0 and −1 if from A_1. A component of the quotient carries a loop of sign −1 exactly when its lift holds some x together with Cx. Where no component does, H_+ and H_- are conjugate by a diagonal sign matrix and have equal spectra. **[proved; RUNS: hexagon and hexagon + terminal, m = 4–14; equal spectra at m = 6, 10 to 2.7×10^{-15}; sealed]** (GPT, § 146; frustration form, § 145)

Proof. A signed graph without a loop of sign −1 is balanced: a gauge s with sign(ab) = s(a)s(b) exists, and it lifts to a labelling of the cover with s(Cx) = −s(x), which separates x from Cx. Conversely a loop of sign −1 lifts to a path from some x to Cx. ■

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For the pivot graph, connected at every width tested and with C free of fixed points, the lift is connected, so the odd loop is forced, not found. GPT's § 145 exhibited one such loop; the frustration test shows that one must exist at every width where the graph is connected.

Theorem 5.3 (the alternating route turns the frame). Start at a = 0101… and flip turns 0, 1, …, m – 1 in order. Every chain on the way is self-avoiding at every m, and the route ends at C(a). In shape it is a closed loop: a single cis dihedral enters at bond 1, walks along the chain and leaves at bond m – 1, and the end has the start's dihedral word. In space the lifted chain ends at its start turned by π about the first bond. **[proved; RUNS: heading pattern m ≤ 257, geometry m ≤ 64, shape m = 6–20; sealed]** (route: GPT, § 145)

Proof. After an even number of flips every heading lies in {5, 0, 1} (in units of 60°), so the chain advances strictly along the first bond; after an odd number the prefix runs in headings {0, 5} and the rest in {4, 5}, strictly below the junction [4, § 146]. The end word C(a) has the start's dihedral word, and Theorem 4.1 turns it into a by the half-turn. ■

The π flux of Theorem 5.2 is therefore the rigid half-turn that a closed loop of shapes leaves on the frame. It is the lattice instance of the gauge structure of deformable bodies, in which a cycle of shape changes rotates the body that performs it [12]. That much is geometry; whether the flux is a physical phase is OPEN.

Proposition 5.4 (propagation in the two sectors). At m = 10 with HHPHPPHHPPHP, γ = 0.7 and the start $(|0000100010\rangle + |C\ 0000100010\rangle)/\sqrt{2}$, evolution in the full space and in the even sector agree to 6.6×10^{-15} at 2,001 times in [0, 10], with 388 amplitudes in place of 776 (6,208 against 12,416 bytes in complex double precision). Under all pivots the contact probability is 0.0017113, 0.0430001 and 0.0335927 at t = 0.5, 1 and 2; it first peaks at $t \approx 1.405$ and reaches 0.10186 at $t \approx 6.255$, its maximum to t = 10. Under hexagon + terminal moves the start lies in a component of two chains, neither with a contact, so its contact probability is zero for all time by support. **[RUNS; sealed]** (integrated manuscript, § 147)

The two-state island is not representative of local dynamics: it is one of 192 components of that graph (Section 6), and the full-space and sector agreement is algebra, equal to round-off whichever solver is used.

Remark 5.5 (a difference, not a gap). At m = 10 with γ = 0.7, the largest difference between the sorted eigenvalues of H_+ and H_- is 0.328473 under back2 and 0.338886 under both2 (the vocabularies of Section 6). Each block's lowest eigenvalue lies below the other's highest, so the spectra overlap and the number is not a gap between them. **[RUNS; sealed]** (Gemini's figure, corrected wording, § 151)

6. The price of locality

The hole. A pivot at turn i moves m – i beads at once, and a real chain moves a few beads at a time. What does locality cost, and where does the cost land?

Proposition 6.1 (the tail budget). At m = 10, allowing only flips of the last k turns leaves 404, 214, 112, 58, 30, 16, 8, 4, 2 and 1 components for k = 1 to 10. No allowed move changes the centre while k < m/2, because the centre is the middle dihedral (Theorem 4.2) and only e(4) and e(5) touch it. With the last four turns free, 88 of the 776 chains can reach a contact. **[RUNS; sealed]**

Proposition 6.2 (the shortest local rewrite). Group the words of ℓ turns by their net displacement and outgoing heading. The first collision is at ℓ = 4, and it is only 0110 against 1001. Replacing one by the other swaps the two three-bond arcs of a hexagon: it moves beads i + 2 and i + 3 and nothing else, keeps the last

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bead and the number of right turns, and is its own inverse. **[RUNS: ℓ = 1–4; every hexagon edge to m = 12; sealed]** (GPT, § 146)

Theorem 6.3 (the gate is geometry). A window at i reads 1001 or 0110 exactly when the dihedrals at bonds i + 1, i + 2, i + 3 read trans, cis, trans, and exactly when bonds i and i + 4 lie on one line. **[proved; RUNS: 12,304 windows, m = 4–10; sealed]** (§ 154)

Proof. 1001 and 0110 are the four-turn words whose adjacent differences are 101. Their middle three bonds form half a hexagon, whose flanking bonds point the same way along the line through beads i + 1 and i + 4. ■

Five move sets recur below. Hexagon: the gated rewrites 1001 ↔ 0110. Hexagon + terminal: add e(m − 1). Back2: add e(m − 2) as well. Both2: the hexagon rewrites with e(0), e(1), e(m − 2), e(m − 1). Both3: the hexagon rewrites with e(0), e(1), e(2), e(m − 3), e(m − 2), e(m − 1). Undo the global rotation and e(0), e(1), e(2) move only the first one, two or three beads, so both2 and both3 are local at both ends.

Theorem 6.4 (the mirror-return ladder). Over GF(2) the hexagon masks span a subspace of codimension 3 in $\{0, 1\}^m$; hexagon + terminal gives codimension 2, back2 gives 1, and both2 gives 0, or 1 when $m \equiv 2 \pmod 4$. The mirror mask 1^m lies in the span exactly when $m \equiv 0 \pmod 4$ for hexagon moves, $m \equiv 0$ or 1 for hexagon + terminal, $m \not\equiv 3$ for back2, and for every m under both2. In each of these classes a legal route leads from a word to its mirror image, and a mirror-stable component exists in exactly these classes. **[span RUNS m = 6–257 by elimination; components RUNS m = 4–14; routes legal at every m ≤ 257; sealed; the back2 and both2 rows are new here]** (hexagon and hexagon + terminal: GPT, § 146)

Necessity is the span: a route to the mirror is a sum of move masks. Sufficiency is a route.

- Hexagon, m = 4k: $(1001)^k \to (0110)^k$ by k disjoint windows. With the terminal, m = 4k + 1: $(1001)^k0$, then e(m − 1). Every intermediate heading lies in {5, 0, 1}, so these routes are legal at every m [4, § 146].
- Back2, m = 4k + 2: $(1001)^k10$, its k windows, then e(m − 1), then e(m − 2).
- Both2, m = 4k + 3: $0(1001)^k10$, its k windows at turns 1, 5, …, then e(0), e(m − 2), e(m − 1).

Proposition 6.5 (the frozen bead). Hexagon moves touch beads 2 to m − 1 and the terminal flip touches bead m + 1 alone, so under hexagon + terminal bead m never moves, and the graph has at least as many components as bead m has positions. Hexagon moves alone keep the last bead and the number of right turns. **[proved; RUNS m = 4–14; sealed]** (§ 146)

GPT's § 146 called the obstruction a two-dimensional hole; it is a bead, and its two dimensions are the bead's x and y. At m = 10 the span leaves 2^2 cosets, bead m has 69 positions and the graph has 192 components (Table 2): the span sees a two-bit shadow of the frozen bead, and the rest is the chain jamming on itself, which no count of masks can see.

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m	chains	hexagon	(last bead, right turns)	hexagon + terminal	positions of bead m	back2	both2
4	16	15	15	7	7	3	1
5	30	26	26	13	13	5	1
6	58	46	44	22	20	9	2
7	112	80	72	38	30	14	3
8	216	141	110	67	41	25	5
9	406	236	160	115	55	43	7
10	776	404	227	192	69	73	14
11	1,472	682	308	328	84	120	23
12	2,796	1,157	397	557	101	207	35
13	5,260	1,926	502	935	119	351	63
14	9,964	3,244	622	1,560	139	587	106

Table 2. Components of the move graphs, m = 4–14, beside the invariants that bound them from below: the (last bead, number of right turns) classes for hexagon moves and the positions of the frozen bead m for hexagon + terminal.

One end was missing. The fixed first bond is a frame, not physics. GPT's § 146 vocabulary moved only the back end, because in the fixed frame a flip of turn 0 or 1 looks like a pivot of the whole chain; undo the global rotation and it moves one bead or two. With both ends free the graph at m = 10 drops from 192 components to 14, and 612 of the 776 chains can reach a contact (Table 3).

m = 10, HHPHPPHHPPHP (776 chains, 32 with a contact)	components	largest	chains that can reach a contact
all pivots	1	776	776
last four turns	58	15	88
hexagon	404	10	42
hexagon + terminal	192	20	44
back2	73	64	64
both2	14	186	612
both3	1	776	776
both2 + two-hexagon exchange	1	776	776

Table 3. The vocabularies at m = 10. Locality costs connectivity; moves at both ends and one more bead of reach recover it at this width.

Prior art owns the rest. Madras and Sokal proved that every Monte Carlo dynamics for the self-avoiding walk built on a finite repertoire of local, length-conserving moves is non-ergodic, and that for long chains each ergodic class is an exponentially small fraction of the space [6]. Whether their argument applies to the honeycomb as written has not been checked here, but every count in Tables 2 and 3 and in Section 7 behaves as the theorem says. "Add two more directions and the local graph reconnects" is therefore a DEAD END as a goal: two directions are necessary, and no finite local alphabet is sufficient. The open question becomes which non-local steps the physics allows and at what cost. Sections 7 and 8 answer its geometric half.

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7. Gates, exchanges and locks

The hole. With both ends free the graph at m = 10 still has 14 components. Some are islands that no local rewrite leaves. What are they, what is the next move, and where does each finite vocabulary stop?

Proposition 7.1 (the odd-aligned gate, and the zigzag family). (a) For a window at odd offset 2j + 1, central-pair parity $p_{j+1} = 0$, with $p_t = b_{2t} \oplus b_{2t+1}$, is necessary for a hexagon move and not sufficient. At m = 10 there are 352 odd-aligned firings, each flipping p_j and p_{j+2} and no other pair parity, while 654 odd windows have central parity 0 but read neither 1001 nor 0110 (0000 among them), and 8 more read correctly but give a self-crossing chain. (b) For even $m \geq 6$ let Z_m be the words whose odd interior bonds 3, 5, …, m − 3 are all trans. Z_m has $2^{(m+4)/2}$ words, every one self-avoiding with no non-bonded contact, so it is contact-free for every sequence; and it is one mirror-stable component of the both2 graph. At m = 4 the statement fails: Z_4 is all 16 words, and 0000 puts bead 0 next to bead 5. **[proved; RUNS: (a) m = 10; (b) contacts m = 6–24, component m = 6–20; sealed]** (Gemini's both2 write-up, with the corrected wording of § 151)

Proof of (b). Let c be the heading after the first pair of turns. Bonds 1 to m − 1 have headings within 60° of c, so each advances at least ½ along c, and only the first and last bonds can point back, by ½. A contact needs five or more bonds between its beads, which forces a separation of at least 2 along c. End moves set both end pairs freely, an even-aligned hexagon move swaps two adjacent interior pairs of opposite type (10 and 01), and the first window, with the front pair set opposite, exchanges the type of the first interior pair; so every interior pattern is reachable, and C preserves every pair parity. ■

Proposition 7.2 (ϕ is invariant but does not count). At $m = 4k + 2$ let ϕ(x) be the XOR of the turns at positions $\equiv 2$ or 3 (mod 4). ϕ is invariant under both2, since each window covers two such positions and no end flip touches one. At m = 6, 10, 14 and 18 both2 has exactly two mirror-stable components, Z_m and one other; the other has ϕ = 0 every time and Z_m has ϕ = k mod 2 (Table 4). So ϕ separates the two at m = 6 and 14 and not at m = 10 and 18: ϕ does not explain why there are two. **[RUNS; sealed; the count of two OPEN]** (Gemini; corrected in § 151)

m	chains	both2 components	mirror-stable components: size (ϕ)
6	58	2	26 (0); Z_6 = 32 (1)
10	776	14	186 (0); Z_{10} = 128 (0)
14	9,964	106	1,018 (0); Z_{14} = 512 (1)
18	125,168	858	5,114 (0); Z_{18} = 2,048 (0)

Table 4. The two mirror-stable both2 components at $m \equiv 2$ (mod 4).

The next move. The honeycomb's simple cycles through a vertex have lengths 6, 10, 12, 14 and so on, never 4 or 8 (checked to 14). The hexagon rewrite is the 6-cycle. The next local exchange is the 10-cycle, two hexagons sharing a bond: replace a five-bond arc of the chain by the other five-bond arc with the same ends and a disjoint interior (Gemini, § 148). Read in the chain's own frame it is symmetric under reversal, and its front window is an end move like any other.

Proposition 7.3 (the two-hexagon exchange). At m = 6, 8, 10, 12 and 14 the symmetric 10-cycle exchange has 20, 108, 508, 2,266 and 9,710 edges. The interior windows give 4, 48, 292, 1,498 and 6,958 of them and never leave Z_m; the one-ended set, which omits the front window, gives 12, 78, 400, 1,882 and 8,334 and leaves Z_m 8, 16, 32, 64 and 128 times, always at the back window i = m − 4; the symmetric set leaves it 16, 32, 64, 128 and 256 times. An interior exchange changes six turns by 101101 or 110011, a front exchange five turns by

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01101 or 10011, a back exchange by 10110 or 11001. Added to both2, the symmetric exchange leaves 1, 3, 1, 3 and 13 components; the one-ended exchange leaves 1, 3, 3, 5 and 17, and at m = 10 it splits 766, 5, 5 where the symmetric graph is whole. **[RUNS; sealed]** (Gemini, § 148)

Gemini's one-ended result is the same frame artifact as the missing front end of § 146.

Remark 7.3a (a contact by four kinds of move). 0000100010 → 0100100010 → 0011000010 → 0111000010 → 1100010010 → 1000010010 is a legal route at m = 10, step by step an end move, a hexagon rewrite, an end move, a 10-cycle exchange and an end move, from a chain with no contact to a chain with one on HHPHPPHHPPHP. **[RUNS; sealed]** (Gemini, § 148) Its start is the word whose hexagon + terminal component has two chains (Proposition 5.4).

Proposition 7.4 (the Ω-loop islands). At m = 8 and 12 the small components of both2 with the symmetric exchange are exactly W_m and $C(W_m)$, where W_{4k} is the set of words {00, 10}·(1111 0000 1111 …, k − 1 blocks)·({00, 01} for k even, {10, 11} for k odd). W_{4k} is valid and closed under both moves for every m = 8 to 32. **[RUNS; sealed]** (Gemini)

The block 1111 is five bonds of a hexagon, which leave its entry and exit adjacent: an Ω-loop, closed by geometry. Its reading as a lock of *One Inverter* is a DEAD END, since $\tau^4(11110000) = 00000000 \neq C(11110000)$ and a lock needs an odd number of lanes [3, Theorem 4.2]. The island belongs to the lattice, not to the ring.

Theorem 7.5 (where three turns at each end stop). The both3 graph on V_m is connected for every m = 4 to 21, where it holds 826,564 chains. At m = 22 to 26 it has 5, 9, 23, 55 and 117 components, and 4, 4, 12, 18 and 42 chains are isolated (Table 5). The symmetric 10-cycle frees all four isolated chains at m = 22 and leaves 4, 4, 10 and 28 locks at m = 23 to 26; the 28 at m = 26 are seven orbits of four under {I, C, J, JC}. **[RUNS: exhaustive; two compiled searches agree; sealed]** (counts: Gemini and GPT, §§ 149–150; exhaustive, §§ 151 and 154)

m	chains	components	outside the largest	isolated	locks after the exchange
21	826,564	1	0	0	0
22	1,551,488	5	4	4	0
23	2,903,404	9	12	4	4
24	5,442,740	23	48	12	4
25	10,175,458	55	162	18	10
26	19,053,856	117	430	42	28

Table 5. The both3 census, exhaustive. "Locks" are the isolated chains that the symmetric two-hexagon exchange does not free.

Proposition 7.6 (the 22-turn clasp). w = 00011110101101101111100 is isolated under both3. The front exchange, at window i = 0, takes it to 01110110101101101111100, which has a both3 move. On the reversed word the same exchange sits at i = 18, and the four-bead end move e(3) puts beads 0 to 3 on the same four sites, (−3, 2), (−3, 3), (−2, 1) and (−1, 1). **[RUNS; sealed]** (Gemini, § 149)

Proposition 7.7 (longer exchanges). Adding 12-cycle and then 14-cycle exchanges to both2 with the symmetric 10-cycle gives 1, 3, 1, 3 and then 1, 2, 1, 1 components at m = 6, 8, 10, 12; at m = 14 the count stays at 13, the largest component holding 9,920 of 9,964 chains. **[RUNS; sealed]** (Gemini's figure; unlinked in § 151, run in § 154)

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Every finite vocabulary tested moves its frozen chains to longer widths, as the Madras–Sokal theorem leads one to expect (Figure 2). The next move is the four-bead end gate, and on the locks the exchanges leave it fails in the plane more often than not (Section 8).

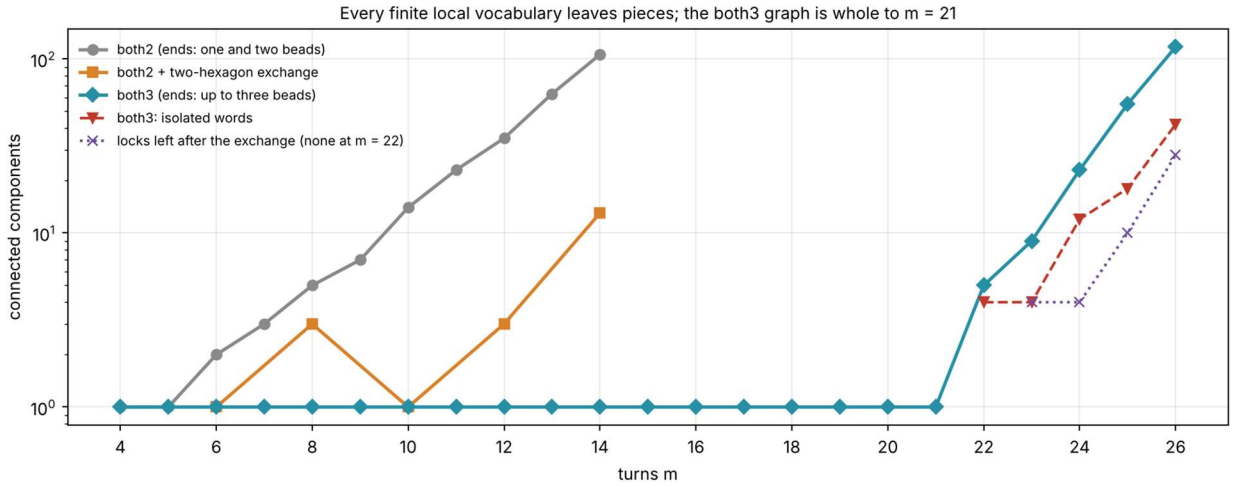


Figure 2. Components of the move graph against width, on a logarithmic scale. Grey: both2. Orange: both2 with the symmetric two-hexagon exchange. Teal: both3, one component from m = 4 to 21 and exhaustive to m = 26 by compiled search. Red: chains isolated under both3. Purple: locks left after the exchange (none at m = 22).

8. Leaving the plane

The hole. The four-bead end gate e(j), j = 3 or m − 4, rotates a block of four beads by 120° about its pivot in the plane, and on the locks the planar arcs sweep the block through the chain (Proposition 8.6). It changes two dihedrals (Proposition 4.3), so in space it is a torsion motion. What is that motion, and is it free?

Bonds here are segments of zero thickness, and the clearance of a configuration is the minimum distance between non-incident bonds.

Theorem 8.1 (half-turns). For a rotation R_θ about an axis that lies in one plane with the points p and q,

$$|R_\theta p - q|^2 = ((1 + \cos\theta)/2)\cdot|p - q|^2 + ((1 - \cos\theta)/2)\cdot|R_\pi p - q|^2.$$

So a rigid half-turn of part of a planar chain about the line of one of its bonds never brings two bonds closer than the smaller of its two endpoint clearances. **[proved; RUNS: 2,000 random cases; on all 160 serial half-turns of the locks the sampled minimum equals the endpoint minimum; sealed]** (GPT, § 152)

A tail mirror D_k and a prefix mirror Q_k are rigid half-turns of this kind, and both of their endpoints are planar words. When both words are valid the motion is collision-free, which sharpens § 151's argument (the moving part meets the plane only on the axis) to an exact minimum.

Theorem 8.2 (the hexagon rewrite is a crankshaft). The hexagon rewrite at window i is the half-turn of beads i + 2 and i + 3 about the line through beads i + 1 and i + 4. Every bond length stays 1 and every bond angle 120°, only the dihedrals at bonds i and i + 4 change, and the motion is collision-free whenever both words are valid. **[proved; RUNS: all 697 hexagon edges, m = 4–10; sealed]** (§ 154)

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Local moves have been the stock of lattice-polymer dynamics since Verdier and Stockmayer [8], and the theorem of [6] is why no finite family of them is ergodic. The two-bead crankshaft is one of them; here it is forced by the gate geometry of Theorem 6.3.

Theorem 8.3 (the end gate is one rigid rotation). Run the end gate at turn j as two torsions, θ_1 about bond j and θ_2 about bond j + 1, together, $(\theta_1, \theta_2) = (\theta, \sigma\theta)$ with $\sigma = \pm 1$ (GPT, § 153). The four end beads and the pivot bead j + 1 move as one rigid body about the pivot, with every bond length and bond angle fixed, at constant angular speed $\sqrt{(2 + \sigma)}$: 1 counter-rotating and $\sqrt{3}$ co-rotating. At $\theta = \pi$ the chain is planar again and reads $x \oplus e(j)$. The in-plane pivot instead drives the bond angle at the pivot through 180° halfway. **[proved; RUNS: 68 gates, both senses; sealed]** (§ 154)

Proof. Both rotations fix the pivot, so the block moves rigidly about it. One axis is fixed in space and the other is carried by the first rotation, which keeps its angle with the fixed axis; consecutive bonds meet at 60°, so with unit axes u_1 and u_2 the angular velocity has $|\omega|^2 = 1 + 1 + 2\sigma\, u_1 \cdot u_2 = 2 + \sigma$. ■

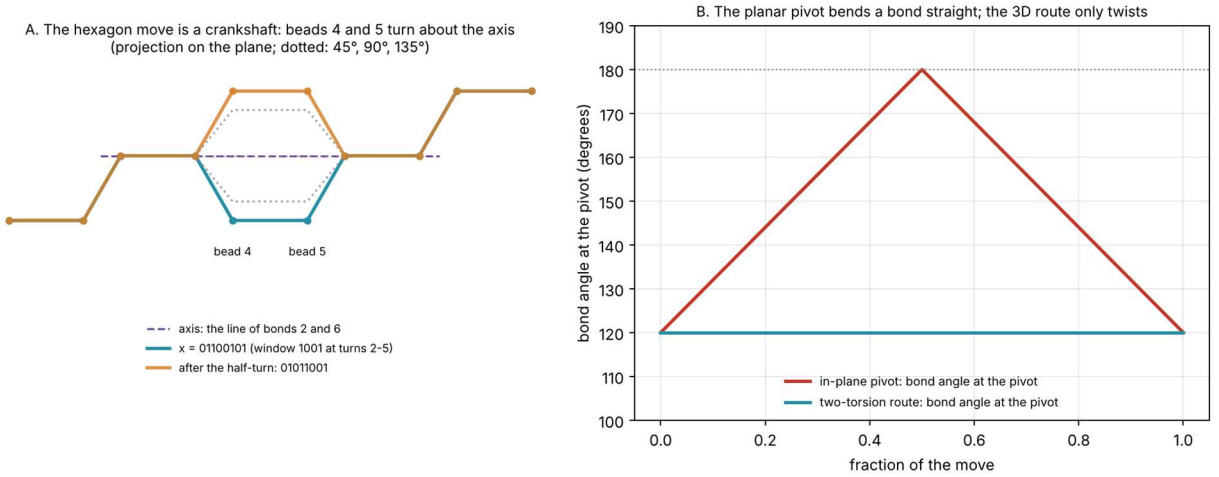


Figure 3. Two torsion motions. (A) The hexagon rewrite 01100101 → 01011001 is a half-turn of beads 4 and 5 about the line of bonds 2 and 6, shown in projection on the plane; dotted, the beads at 45°, 90° and 135°. (B) The bond angle at the pivot during gate 3 of x*: the in-plane pivot bends it to 180° halfway, and the counter-rotating torsion route keeps it at 120° throughout.

Proposition 8.4 (serial routes). Since $e(j) = D_j \oplus D_{j+1} = Q_{j-1} \oplus Q_j$, an end gate can also run as two half-turns in series, through a corner word $x \oplus D_j$ or $x \oplus D_{j+1}$ (Q for the front), each collision-free by Theorem 8.1 when the corner is valid. The 46 locks have 68 endpoint-valid four-bead gates (turns 3 and m − 4). 56 of them have a valid corner and hence a serial route, every lock has at least one, and 12 have both corners self-intersecting; at m = 26, 28 of 36 gates have a serial route, with 44 serial orders between them. **[RUNS; sealed]** (route: § 151; count corrected in § 153: 68 = 56 + 12)

Theorem 8.5 (the synchronous certificate). Along $(\theta, \sigma\theta)$ each moving bead travels at most $\sqrt{(2 + \sigma)} \cdot R$ per radian, R its fixed distance from the pivot, and the clearance is 1-Lipschitz in the bead positions. With 360 samples on $\theta \in [0, \pi]$, the sampled minimum less $\sqrt{(2 + \sigma)} \cdot R_{max} \cdot \pi/720$ is positive for at least one σ on every one of the 68 gates, and its least value over the gates is 0.977328. Every end gate on every lock, including the 12 whose serial corners are blocked, is therefore collision-free along a synchronous path. Over all gates the sampled minima are 0.912897 co-rotating and 0.654657 counter-rotating. **[proved bound; RUNS; sealed]** (path and certificate: GPT, § 153; exact constant: § 154)

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Gemini's tighter constant, $r_1 + r_2$ from the planar distances of the beads to the two axes, does not bound the speed. The first rotation changes each bead's distance to the second axis, and on 8 of the 72 gate-directions at m = 26 the true speed exceeds $r_1 + r_2$, by up to 1.5 times. That derivation is a DEAD END; the exact constant gives a valid and stronger certificate.

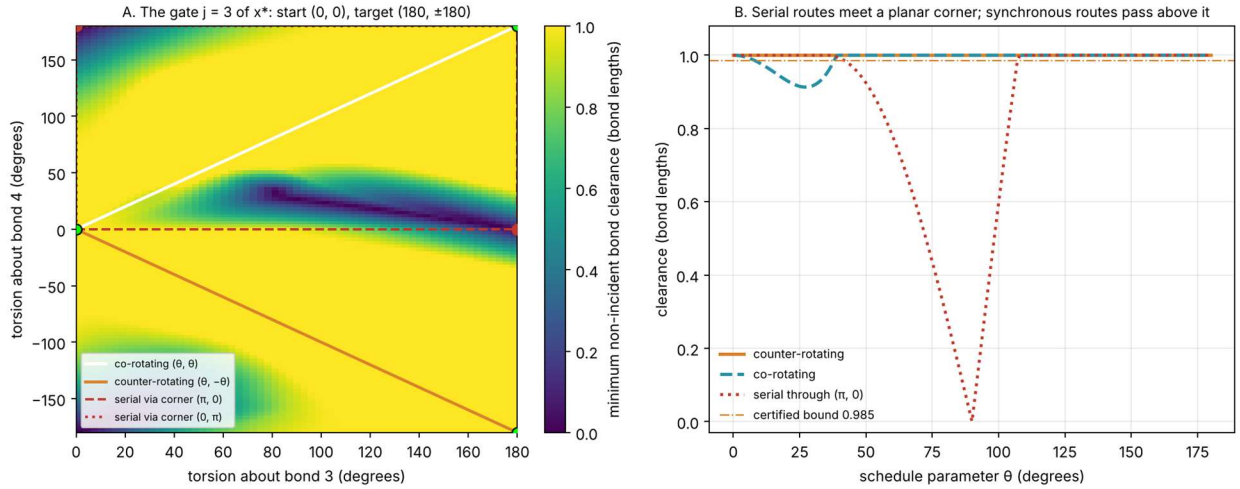


Figure 4. The control torus of gate 3 on x^*. (A) Clearance over the torsion θ_1 about bond 3 (0° to 180°) and θ_2 about bond 4 (−180° to 180°). The start (0, 0) and the targets (180°, ±180°) are green; the serial corners (180°, 0) and (0, 180°) are planar words that cross themselves (red). (B) Clearance along three routes. The serial route through (180°, 0) reaches clearance 0 at the corner; the counter-rotating route stays at 1, with certified bound 0.985; the co-rotating route dips to 0.913.

Proposition 8.6 (planar against spatial). A certified clear in-plane arc exists for 4 of the 4 locks at m = 23, 0 of 4 at m = 24, 4 of 10 at m = 25 and 0 of 28 at m = 26, and every uncertified arc has a sampled collision. **[RUNS; sealed]** (§§ 151 and 154) GPT's § 150 found the four-bead gate open in the plane on three clasps (Proposition 8.6a) and closed on 3 of the 7 orbits at m = 26; on the full set of locks the plane opens 8 of 46 and space opens all 46 (Theorem 8.5).

Proposition 8.6a (the clasps of § 150). For 00011110101101101111 00, 00001000101001001000011 and 00011110110110010010000111 the end move e(3) lands on a valid word, and the in-plane 120° pivot of beads 0 to 3 about bead 4 is certified clear, with sampled minima 1.000, 0.645756 and 1.000 and certified bounds 0.997091, 0.642847 and 0.997091 (720 samples). **[RUNS; sealed]** (GPT, § 150)

On the locks the hole was a missing degree of freedom, the one the planar model removed, not a missing rewrite.

Proposition 8.7 (chirality). Let F(x; ϕ) be the chain x after the torsions ϕ. Then $R_{\pi,x}F(x; \phi) = F(Cx; \phi)$ and $P_z F(x; \phi) = F(x; -\phi)$, to machine zero. **[proved; RUNS: 12 gates, random torsions; sealed]** (Gemini) The planar frame flip is a rotation, and the mirror of space negates the torsions: a chain with a handed torsion preference would tell x from C(x) only through the sign of its twists.

9. What is compiled, and what is not

The hole. The word compiles the chain's geometry, its symmetries and the shapes of its moves. A quantum reading asks for more: amplitudes, rates, phases and an environment. What does the construction supply, and what must come from outside?

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Proposition 9.1 (the projector block law). If two chains lie in different components of the move graph, every function of H, e^{-iHt} included, has zero amplitude between them. At m = 10 under both2 (14 components) the largest such amplitude at t = 1, 5 and 10 is 1.7×10^{-14}, which is round-off. **[proved; RUNS; sealed]**

Rates and phases cannot bridge a missing edge, and a quantum walk on the move graph explores exactly what the moves connect. The third dimension is therefore not a quantum effect but a degree of freedom: no choice of amplitudes on the planar graph could have opened the locks. What the sector structure adds is exact bookkeeping (half the amplitudes, a π flux) and a holonomy that is a rigid half-turn.

Proposition 9.2 (the relation layer). On the tree with edges 01, 12, 13, 34, 35, 56, let K tokens slide along edges onto empty vertices, and let each pair of moves on disjoint edges fill a square. At K = 2 the complex has 21 configurations, 30 moves and 8 squares, its boundary has rank 8 over $\mathbb{Q}$, and $\beta_1 = 2$; at K = 3 the counts are 35, 60, 24, rank 22 and $\beta_1 = 4$. The unsigned incidence matrix has rank 24 at K = 3, so it is not a boundary operator. **[RUNS; sealed]** (Gemini, § 149; corrected in § 151)

β_1 counts the cycles of moves that independent moves do not fill. On the chain it is the measure of the route loops that no commuting pair explains, the place where a loop of shapes can carry a holonomy.

Prior art.

- Turn encodings of lattice proteins on qubits are established. Perdomo-Ortiz et al. encode two bits per relative bond direction and anneal [9]; Robert et al. use a tetrahedral lattice and a variational eigensolver [10]. Both compute lowest-energy folds, not kinetics. Here the encoding is one bit per turn, and the content is in the moves, their sectors and their geometry.
- The pivot graph is the pivot algorithm [5, 7], and the impossibility of a finite ergodic local move set is a theorem [6]. Local moves in lattice dynamics go back to Verdier and Stockmayer [8].
- The loop τ is the twisted translation of a spin ring [11], and the frame holonomy is the gauge kinematics of a deformable body [12].

What this paper does not claim

Everything here is the combinatorics and geometry of lattice chains with unit bonds, fixed 120° angles and bonds of zero thickness. No bead size, torsion barrier, rate, solvent or temperature enters, and no folding mechanism is derived. No speed-up or measured power is claimed. H_- and the frame holonomy have no physical reading here. The gate counts per Strang step reported in the integrated manuscript of § 147 were not reproduced and are not used. Nothing here bears on any hash function.

The next stroke

Six routes are open, each with its missing link named.

1. **The pivot graph at every width.** It is connected to m = 14. The missing link is a matching of Madras and Sokal's ergodicity proof [7] to the honeycomb and the two 120° pivots.
2. **Two mirror-stable components.** Both2 has exactly two at m = 6, 10, 14 and 18, and ϕ does not explain the count. The missing link is the second invariant.
3. **Every lock in space.** All 46 locks at m = 23 to 26 open through a four-bead gate along a synchronous path. The missing link is a clearance bound for that path that does not depend on the lock.
4. **The cost of the third dimension.** Thickness, torsion barriers, rates and solvent decide whether the crankshaft and the twisted end gate are cheap enough to be the moves a real chain uses.

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5. **The odd sector.** A physical reading of H_- and of the frame holonomy, if one exists.
6. **The honeycomb in the non-ergodicity theorem.** Whether Madras and Sokal's 1987 argument covers the honeycomb as stated [6].

Conclusion

A turn word is a chain. Its adjacent differences are the chain's shape and its one missing bit is the frame, so the mirror is a half-turn of space and every move is a set of torsion flips. A Hamiltonian that respects the mirror splits in two, and its odd half carries a π flux exactly where a component reaches its own mirror image; the flux is the half-turn that a closed loop of shapes leaves on the frame. Locality costs connectivity in a way the mask algebra counts only in part: a span that admits the mirror at some widths and not at others, a bead that never moves, and the theorem, borne out at every width tested here, that no finite set of local moves reaches every self-avoiding walk. With three turns free at each end the plane holds together to 21 turns and then leaves locks, and every lock found opens once the chain may twist: a four-bead end block turns rigidly about its pivot, clear of the chain by most of a bond length at every instant. The way between two planar states was never in the plane.

Acknowledgment. Computation and drafting were done with Claude (Anthropic) as a working partner, which also led the audit of the other runs. Several results were first reached in independent AI-assisted runs on 23 October and are recorded with their seals in the ledger [4, §§ 144–154]: ChatGPT (the six-ring constraint audit, the honeycomb route compiler and its two-sheet reduction, the locality analysis with the hexagon rewrite and the mod-4 law, the in-plane end-gate geometry, the half-turn identity and the synchronous two-torsion certificate); Gemini (the both2 component analysis with Z_m and ϕ, the two-hexagon exchange and its witness route, the Ω-loop islands, the relation layer, the 22-turn clasp, the mirror identity $e(j) = D_j \oplus D_{j+1}$, the 46-lock audit with chirality, and the control torus). The integrated manuscript audited in § 147 contributed the propagation run of Proposition 5.4, and a pasted five-point review corrected Gemini's theorem block. Every number in this paper was recomputed by the notebook of Appendix C.

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Appendix A. Ledger

Claim	Status	Scope
2.1–2.4 six rings: dominoes, no move; leakage 49/75; two paths; bridges; penalties; Strang; HPPH	proved; RUNS; sealed	GPT § 144, § 147
2.5 a merge leaves an orthonormal environment record	proved; RUNS; sealed	SWAP, $\mathbb{C}^6$
2.6 $U_\tau = X_{last} \cdot R$, $U_\tau^n = X^{\otimes n}$, spectrum by orbits	proved; RUNS; sealed	n = 2–8
3.1 encoding: C mirror, JC reversal, flip = 120° pivot; $\|V_m\|$	proved; RUNS; sealed	m ≤ 14
3.2 pivot graph connected	RUNS; sealed	m = 2–14; all widths OPEN
3.3 C keeps every energy; J, P change 48 of 776	proved; RUNS; sealed	m = 10
4.1 shape = dihedral word; classes {x, Cx}; C = half-turn	proved; RUNS; sealed	32,764 words; 5,894 chains
4.2 (dihedral word, face) returns x; centre = middle dihedral	proved; RUNS; sealed	n = 2–16
4.3 moves as torsion flips; $e(j) = D_j \oplus D_{j+1}$	proved; RUNS; sealed	m = 6–64
5.1 $H = H_+ \oplus H_-$; multigraph quotient	proved; RUNS; sealed	m = 4–8
5.2 odd loop iff the lift holds x and Cx; equal spectra otherwise	proved; RUNS; sealed	m = 4–14
5.3 alternating route legal at every width; frame half-turn	proved; RUNS; sealed	m ≤ 257
5.4 propagation; contact values; two-state island	RUNS; sealed	m = 10
5.5 sorted-eigenvalue difference, not a gap	RUNS; sealed	m = 10
6.1 tail budget; centre fixed while k < m/2	RUNS; sealed	m = 10
6.2 shortest rewrite 1001 ↔ 0110	RUNS; sealed	ℓ ≤ 4; m ≤ 12
6.3 gate ⟺ trans–cis–trans ⟺ collinear flanking bonds	proved; RUNS; sealed	12,304 windows
6.4 codimensions; mirror-return classes; routes; mirror-stable components	STANDS (span); RUNS; sealed; back2, both2 new	m ≤ 257; m = 4–14
6.5 frozen bead; Tables 2 and 3	proved; RUNS; sealed	m = 4–14
a finite local move set that connects all chains	DEAD END as a goal (Madras–Sokal 1987)	—
7.1 odd gate necessary, not sufficient; Z_m	proved; RUNS; sealed	m = 6–24
7.2 ϕ invariant; two mirror-stable components	RUNS; sealed; the count OPEN	m = 6–18
7.3 cycles 6, 10, 12, 14; 10-cycle catalogue; § 148 route	RUNS; sealed	m = 6–14
7.4 Ω-loop islands W_m	RUNS; sealed	m = 8–32
7.4 Ω-loop as a *One Inverter* lock	DEAD END (q = 2 is even)	—
7.5 both3 connected to m = 21; census m = 22–26; 46 locks	RUNS (exhaustive, compiled); sealed	m ≤ 26
7.6 the 22-turn clasp	RUNS; sealed	m = 22
7.7 12- and 14-cycles	RUNS; sealed	m = 6–14
8.1 half-turn identity; serial sweep minimum = endpoint minimum	proved; RUNS; sealed	160 half-turns
8.2 hexagon rewrite = crankshaft, collision-free	proved; RUNS; sealed	697 edges
8.3 end gate = rigid rotation at $\sqrt{(2+\sigma)}$; planar pivot bends to 180°	proved; RUNS; sealed	68 gates
8.4 serial routes: 56 of 68 gates, every lock	RUNS; sealed	m = 23–26

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Claim	Status	Scope
8.5 every gate certified at 0.977328 or better	proved bound; RUNS; sealed	68 gates
Gemini's $r_1 + r_2$ constant	DEAD END (fails on 8 of 72)	m = 26
8.6 clear planar arcs 4/4, 0/4, 4/10, 0/28; the § 150 clasps	RUNS; sealed	46 locks
8.7 chirality	proved; RUNS; sealed	12 gates
9.1 projector block law	proved; RUNS; sealed	m = 10
9.2 relation layer $\beta_1 = 2, 4$	RUNS; sealed	K = 2, 3
the ring as the physical mechanism of folding; rates, thickness, solvent; a reading of H_-	OPEN	—

Table 6. Ledger. "Sealed" marks the statements hashed in § 155 before the notebook first ran; most were first sealed and run in §§ 144–154.

Corrections logged on the way

The rotation. § 144's first draft rotated by (a, b) ↦ (−b, a − b), which is 60°; the 120° rotation in the 0°/60° basis is (a, b) ↦ (−a − b, a). Fixed before the first run.

Solver digits. The integrated manuscript (§ 147) quoted the full-versus-sector gap as below 5×10^{-16}. That is solver-dependent (eigendecomposition gave 1.7×10^{-15}, a sealed threshold that failed; its own solver gave 4.4×10^{-16}); the equality is algebra. Its Figure 7 compared the full graph with a two-state island and hid the first peak at $t \approx 1.405$. Its reference [1] carried a hash-function section that this series keeps out; *One Inverter* supplies the ring mathematics.

Five review points (§ 151). The odd gate is necessary, not "iff"; two mirror-stable components are not two ϕ sectors; "contact-free at every even width" fails at m = 4; the "spectral gap" is a sorted-eigenvalue difference; $\Lambda_2(1111)$ is not a *One Inverter* lock.

Twelve, then twenty-eight. § 150's twelve residual 26-turn locks were 3 of 7 orbits; the exhaustive count is 28.

The relation layer. An unsigned incidence matrix is not a boundary operator over $\mathbb{Q}$ (rank 24, not 22); the corrected β_1 are 2 and 4.

The front window. Gemini's one-ended 10-cycle omitted the front window, the same frame artifact as § 146's missing front end; with it, m = 10 is one component.

Unlinked bars. The 12- and 14-cycle bars at m = 14 in Gemini's figure had no run behind them in § 151; § 154 ran them (13 components).

§ 153's own corrections. 68 = 56 + 12, not "68 matches 56"; "68/68 counter-rotating" took the better of two senses per gate; front gates are compared in turn words, not lab frames; its Lipschitz constant 14 is valid but loose.

The tightened constant. Gemini's $r_1 + r_2$ fails on 8 of 72 gate-directions, so its 0.9093 and 0.9395 do not follow; its sampled minima 0.913132 and 0.654730 become 0.912897 and 0.654657 on a finer grid.

This paper's notebook. Before its first run the route check was made to require the gate (1001 or 0110) at every window flip, eigen-angles in check 2.6 are rounded before sorting, the parallel-edge check tests existence as sealed, a probability that is zero by support is tested against 10^{-20}, and a list of the 46 locks was added for machines without a C compiler, which the compiled search must reproduce. The first run passed all 44 checks and stopped in the figure cell on a reused variable name. Figure 3A had been drawn on a window that does not read 1001 or 0110; it was redrawn on one that does, and the notebook was rerun in full. The six addendum statements (A1–A6) were hashed after the notebook was drafted and before it first ran.

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Appendix B. Sealed predictions

Each prediction was written with the reading it predicts and the readout it returns through, and its text was hashed before the code that tests it ran. The digests of the GPT and Gemini runs are their own, printed by their notebooks and reproduced in the ledger [4].

Prediction	SHA-256	Result
§ 144 six rings, R_3, U_τ, route encoding	5e1934fda2ca701442616b19c1d96a70 06361caee513e08acbc68e960c9fc4bc	RUNS
GPT § 145 route compiler, geometry	f2c8e0441f278aa4b7fdefa47f9403f0 873731ce02fbe8e2ff4f407241ef4250	RUNS
GPT § 145 two-sheet reduction	7f4ffd691dbff3cf8f9d1b918328f105 2c93b40fb9d6ce8c8297dfe0b25254f8	RUNS
GPT § 146 locality compiler	a2046d768346cdcdb4446afb2dfb7ca2 312b87c49b6b0f3c11a50e1196a3e900	RUNS
GPT § 146 holonomy controls	63cbcfa525538c512b54355fab8837ff c92cbfc1398514d55eb0b97722a9f421	RUNS
§ 146 routes, frozen bead, ladder	a8765b92c2806656d9ba44de21e45ce0 57aa1aaafca3b7feac33f52be575ad37	RUNS
§ 147 integrated manuscript	26c2694d80664c10415375d66ae051ef 2e9e0a5f95741ce36589ece82cc5e2ad	RUNS; one solver threshold failed
Gemini § 148 10-cycle exchange	00137ca45664f8039f645bb37d1fd959 067bd98fc93b3a31b3652b2bf45c76ff	RUNS (rerun § 154)
§ 149 relation-layer script	3cc25cece66ad95a8c355b99455d610a a7a50e3f754d2e001770aed56ff10407	RUNS; corrected
GPT § 150 end-gate geometry	0e03f634d7bc7bb0894a3366ad539c6c cf9ba51f96631f9b160858498b1dc9f9	RUNS
§ 151 review, census, 3D route	0b330f6745b67c2a5ff522882316706e cafe6a7282fc62c818cbe3649b5ebadb	RUNS
GPT § 152 gauge and mirror compiler	997412f0c6ab7ba01f33f143c3101b3c dd81653fa79da01ae9156b34332f01ca	RUNS
GPT § 153 synchronous certificate	6ec4e001b3c5cdd728dc63da20600838 875f97849bd2c281df8161484d5f0861	RUNS
§ 154 dihedral word, rigid rotation, certificate	f54d4d85b9a2fae8890a196364a0319c 16de596d212242c95e02124e853f6374	RUNS
§ 154B both3 census, 10-cycle catalogue	25bc8532b57f683a6f4f7d56ed32fec0 b2edba60b1fe3206b552c9a4eb2a61d3	RUNS
§ 155 this paper	a88c0432ea74d31b11fc5fd6005ca1d9 3a71a57a84b3e4647315f7dc5c259a7a	RUNS, all 38 statements
§ 155 addendum A1–A6	daac3a89e8bb2e393ca09a13bfeb67d9 1ac5f4c8f075bc5d4e3b0cff36c6616f	RUNS, all six

Table 7. Seals. Digests are SHA-256 of the literal prediction text.

The paper's own seal lists every numbered statement with the value it must reproduce, and the addendum adds six statements from the ledger that the paper also uses (A1 is Remark 5.5, A2 is Proposition 7.1(a), A3 is Table 4, A4 is Proposition 8.6a, A5 is Table 2, A6 is Remark 7.3a). New in the paper seal were the back2 and both2 rows of Theorem 6.4, the existence of mirror-stable components by class, the measured route orders, the both3 row of Table 3, and the readings of x^*. The full texts follow as hashed.

```
NEXUS § 155 — The Way Between — paper seal — 2026-10-23 — Dean A. Kulik, QuHarmonics Research Group, ORCID 0009-0003-3128-8828.
Every numbered statement of the paper with the value it must reproduce. §§ 144–154 sealed and ran each of these in the ledger except where marked NEW. This text is hashed before the notebook code is written; the notebook stops if it changes.
```

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2.1 The closed six-step square walks with the first bond east are six 2×1 dominoes; PHPPHP energies −1, −1, 0, 0, 0, 0; pairs differ at 3 sites (6 pairs) or 4 (9); no one-residue move joins two of them.
2.2 The toy's address order is R_3 = 0, 1, 2, 6, 3, 5, 4, 7; its successor T joins words at Hamming distances 1, 2, 1, 2, 2, 1, 2, 3.
2.3 Uniform start on the six valid addresses, K = −0.7(T + T⁻¹): leak coefficient 49/75; ‖QKP‖ = √2·0.7; the valid graph is two paths; projected contact ≤ 1/2, 0.463631173441 at t = 1; virtual bridges 0.49 on addresses 0, 2, 4, 6 and pairs (0, 6), (2, 4).
2.4 Leak at t = 1 for penalties 0, 10, 50, 100: 0.324402465, 0.006480472, 0.000151516, 0.000023742; projected mixer 0. Commutator norms at penalty 10: 10.406248, 109.882164, 24.111397. Strang errors r = 4..128: 0.082770330, 0.018193978, 0.004416348, 0.001096147, 0.000273545, 0.000068356. HPPH contact at t = 1: 0.7717491215.
2.5 A unitary sending six orthogonal states to one output keeps six orthonormal environment records (SWAP).
2.6 U_τ = X_last·R and U_τ^n = X^⊗n; spectrum: one full set of L-th roots of unity per τ-orbit of length L (n = 2..8).
3.1 Honeycomb encoding: C = mirror, JC = reversal (every word to 14 turns); one-bit flip = 120° pivot of the tail (to 11); |V_m| = 2, 4, 8, 16, 30, 58, 112, 216, 406, 776, 1472, 2796, 5260, 9964 (m = 1..14).
3.2 The legal one-bit pivot graph is connected for m = 2..14; edges 3,352 at m = 10 and 57,236 at m = 14.
3.3 m = 10, HHPHPPHHPPHP: C keeps every HP energy; J and P change 48 of 776.
4.1 The dihedral about bond i is cis iff b_(i−1) = b_i (every word, m ≤ 14); valid chains are congruent iff their dihedral words agree, classes {x, Cx} (m ≤ 12, exact); C sends site (a, b) to (a + b, −b), the proper half-turn about the first bond (m ≤ 14).
4.2 face(Cx) = 1 − face(x) (n ≤ 16); centre = dihedral at the middle bond.
4.3 Turn flip e(j): dihedrals j, j+1; hexagon at i: i, i+4; tail mirror D_k: k; prefix mirror Q_(k−1): k (m ≤ 64).
5.1 [H, U_C] = 0, and the C-parity basis splits H into H₊ ⊕ H₋ with residual < 1e-12: pivot graph m = 6, 8; hexagon and hexagon + terminal multigraph quotients m = 4, 5, 6, 8; a cross-sheet self-loop at m = 4 (hexagon), parallel same/opposite-sheet edges at m = 5 (hexagon + terminal) and m = 8.
5.2 A quotient component is frustrated iff its lift holds x and Cx (hexagon, hexagon + terminal, m = 4..14); where none does, H₊ and H₋ have equal spectra.
5.3 The alternating route 0101… → its complement by prefix flips is legal at every width (heading pattern to m = 257, geometry to 64), closes in dihedral words, and ends at the start turned by π about the first bond (m = 6..20).
5.4 m = 10, HHPHPPHHPPHP, γ = 0.7, start (|0000100010⟩ + |C⟩)/√2: full and even-sector propagation agree to < 1e-13 for t ∈ [0, 10]; all-pivot contact 0.0017113, 0.0430001, 0.0335927 at t = 0.5, 1, 2, first peak near t = 1.405, maximum 0.10186 near t = 6.255; under hexagon + terminal the start's component has 2 states and no contact.
6.1 Tail budget k = 1..10 at m = 10: components 404, 214, 112, 58, 30, 16, 8, 4, 2, 1; centre-changing edges 0 for k < 5.
6.2 Heading-and-endpoint signatures collide first at 4 turns, only 0110/1001; the rewrite moves beads i+2, i+3 only and keeps the endpoint and the right-turn count (m ≤ 12).
6.3 Window 1001/0110 ⇔ dihedrals trans–cis–trans ⇔ bonds i and i+4 on one line (m ≤ 10).
6.4 GF(2) codimensions (m = 6..257): hexagon 3, + terminal 2, + bit m−2 (back2) 1, both ends (both2) 0, or 1 when m ≡ 2 (mod 4). 1^m lies in the span iff m ≡ 0 (hexagon), m ≡ 0, 1 (+ terminal), m ≢ 3 (back2, NEW), always (both2, NEW). A mirror-stable component exists exactly in those classes (m = 4..14, NEW). MEASURE: an order of the flips that is legal at every m ≤ 257 for back2 at m = 4k + 2 and both2 at m = 4k + 3 (NEW).
6.5 Under hexagon + terminal bead m never moves (m = 4..14); at m = 10: 192 components against 69 positions of bead m. Vocabulary table at m = 10 (components / largest / states that reach a contact): all pivots 1/776/776; last four turns 58/15/88; hexagon 404/10/42; hexagon + terminal 192/20/44; back2 73/64/64; both2 14/186/612; both3 1/776/776 (NEW row); both2 + symmetric 10-cycle 1/776/776.
7.1 The odd-aligned gate: central parity 0 necessary, not sufficient (m = 10: 654 + 8 counter-windows). Z_m (odd interior bonds trans) is one mirror-stable both2 component of 2^((m+4)/2) words, self-avoiding with no non-bonded contact, for every even m ≥ 6 (contacts m ≤ 24, component m ≤ 20); m = 4 fails (0000).
7.2 m = 6, 10, 14, 18: two mirror-stable both2 components, Z_m and one with ϕ = 0; Z_m has ϕ = k mod 2 (m = 4k + 2).
7.3 The honeycomb's simple cycles through a vertex have lengths 6, 10, 12, 14, … (none of length 4 or 8; checked to 14). 10-cycle catalogue: edges interior 4, 48, 292, 1498, 6958 (no Z exits); one-ended 12, 78, 400, 1882, 8334 (exits 8..128 at i = m − 4); symmetric 20, 108, 508, 2266, 9710 (exits 16..256); components one-ended 1, 3, 3, 5, 17, symmetric 1, 3, 1, 3, 13; masks 101101/110011, 01101/10011, 10110/11001.
7.4 The small symmetric components at m = 8, 12 are W_m, C(W_m), closed for m = 8..32; τ⁴(11110000) = 00000000 ≠ C(11110000).
7.5 both3 connects V_m for m = 4..21 (|V_21| = 826,564); m = 22..26: components 5, 9, 23, 55, 117, isolated 4, 4, 12, 18, 42 (two independent compiled searches agree); after the symmetric 10-cycle 0, 4, 4, 10, 28 remain; at m = 26 seven orbits of four.
7.6 The 22-turn clasp 0001111010110110111100: isolated under both3; the i = 0 exchange gives 0111011010110110111100, which has a both3 move; on its reversal the exchange sits at i = 18; e(3) fills the same four sites.
7.7 Adding 12- and 14-cycles: m = 6, 8, 10, 12 give 1, 3, 1, 3 then 1, 2, 1, 1 components; m = 14 stays at 13.
8.1 |R_θ p − q|² = ((1+cos θ)/2)|p − q|² + ((1−cos θ)/2)|R_π p − q|²; on every serial half-turn of the locks the sweep minimum equals the endpoint minimum.
8.2 Every hexagon edge (m ≤ 10, 697 edges) is a crankshaft half-turn: lengths 1 and angles 120° throughout, only dihedrals i, i+4 move, sweep minimum = endpoint minimum.
8.3 Along (θ, σθ) the end block turns rigidly about the pivot with angular speed √(2+σ), lengths and angles fixed; every in-plane 120° pivot reaches a straight angle at the pivot (68 gates).
8.4 Two-mirror serial routes exist for 56 of the 68 endpoint-valid four-bead gates on the 46 locks (m = 23..26) and for every lock; the other 12 gates have both corners self-intersecting; at m = 26: 28 of 36 gates, 44 serial orders.
8.5 Certified synchronous clearance (constant √(2+σ)·R_max, 360 samples) ≥ 0.977 on all 68 gates; sampled minima co-rotating ≈ 0.9129, counter-rotating ≈ 0.6547; Gemini's $r_1 + r_2$ lies below the true bead speed on 8 of 72 gate-directions at m = 26.
8.6 Certified clear in-plane arc: 4/4 locks at m = 23, 0/4 at 24, 4/10 at 25, 0/28 at 26; every uncertified arc has a sampled collision.
8.7 R_(π,x)F(x; ϕ) = F(Cx; ϕ) and P_z F(x; ϕ) = F(x; −ϕ).
9.1 m = 10 both2: 14 components; exp(−iHt) has no amplitude between them (< 1e-13).
9.2 Tree 01, 12, 13, 34, 35, 56: K = 2: 21, 30, 8, rank 8, β1 = 2; K = 3: 35, 60, 24, rank 22, β1 = 4; unsigned rank 24.
T1 Running example x* = 0101111010110110111011110 (m = 26): valid, isolated under both3, no 10-cycle move; both four-bead gates endpoint-valid; no clear in-plane arc; gate j = 3 has both serial corners self-intersecting, gate j = 22 a serial order; gate j = 3 certified synchronously. MEASURE its dihedral word, face, sampled clearances and certified bounds.

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```
Way home: every move is its own XOR mask, arc swap or half-turn; dihedral words decode with the face bit; the compiled
searches run from source written by the notebook.


NEXUS § 155 — The Way Between — paper seal, addendum — 2026-10-23 — Dean A. Kulik, QuHarmonics Research Group, ORCID
0009-0003-3128-8828.
Written after reading the last unread attachment of the § 152–153 message (a31c0cd6: Gemini's session reading § 151 and
rerunning § 152) and after nb_src.py was drafted, before it was first run. Statements the paper adds to seal_155.txt,
each with the value it must reproduce. All come from §§ 146–151 of the ledger; none is new.


A1 (§ 151 G4). m = 10, HHPHPPHHPPHP, γ = 0.7, H± = E − γ(A0 ± A1) on the 388 mirror pairs: the maximum sorted-eigenvalue
difference between H+ and H− is 0.328473 under back2 and 0.338886 under both2, and each block's lowest eigenvalue lies
below the other's highest. The spectra overlap; the number is not a gap.
A2 (§ 151 G1). m = 10: 352 odd-aligned firings, and each flips the pair parities p_j and p_(j+2) only (p_t = b_2t ⊕
b_2t+1).
A3 (§ 151 G3). C-stable both2 components (size, ϕ): m = 6: 26 (0) and Z_6 = 32 (1); m = 10: 186 (0) and Z = 128 (0); m =
14: 1,018 (0) and Z = 512 (1); m = 18: 5,114 (0) and Z = 2,048 (0). All both2 components: 2, 14, 106, 858; |V_18| =
125,168.
A4 (§ 150, reproduced in § 151). The three § 150 clasps 0001111010110110111100, 0000100010100100100011 and
0001111011011010010000111: the four-bead end move e(3) lands on a valid word, and the in-plane 120° pivot of beads 0..3
about bead 4 that reaches it is certified clear; sampled minima 1.0, at most 0.64576549 (and above it by less than the
interpolation bound 0.0087), and 1.0.
A5 (§ 146). Components m = 4..14: both2 1, 1, 2, 3, 5, 7, 14, 23, 35, 63, 106; back2 3, 5, 9, 14, 25, 43, 73, 120, 207,
351, 587; hexagon + terminal 7, 13, 22, 38, 67, 115, 192, 328, 557, 935, 1560 against 7, 13, 20, 30, 41, 55, 69, 84, 101,
119, 139 positions of bead m; hexagon 15, 26, 46, 80, 141, 236, 404, 682, 1157, 1926, 3244 against 15, 26, 44, 72, 110,
160, 227, 308, 397, 502, 622 classes of (last bead, number of right turns).
A6 (§ 148, checked in § 151 G7). 0000100010 → 0100100010 → 0011000010 → 0111000010 → 1100010010 → 1000010010 is a legal
route (end, hexagon, end, 10-cycle, end) from a word with no contact to a word with one (HHPHPPHHPPHP energy 0 → −1).


Way home: as in seal_155.txt; each statement is recomputed from the turn words by code in the notebook.
```

Appendix C. The verification notebook

NEXUS_155_the_way_between_verification.ipynb is one self-contained notebook: a seal cell that stops if either text above does not hash to its recorded digest, one library cell, one cell for each section of the paper, a cell for Table 1, a figure cell, and a ledger cell that stops unless every check passed. It uses NumPy, SciPy and matplotlib, and a C compiler for the two exhaustive searches of Theorem 7.5, whose source the notebook writes itself; without a compiler the census is skipped and the 46 locks it lists are re-checked one by one. Run in a fresh kernel it takes about three minutes. Its ledger, unedited:

```
2.1   PASS  closed six-step square walks: six 2x1 dominoes, energies -1,-1,0,0,0,0, no one-residue move between them
[every walk with the first bond east]
2.2   PASS  the toy address order is R_3 and its successor T joins words at Hamming distances 1,2,1,2,2,1,2,3  [R_3 = [0,
1, 2, 6, 3, 5, 4, 7]]
2.3   PASS  leak coefficient 49/75, |QKP| = sqrt2 gamma, valid graph = two paths, projected contact <= 1/2 (0.463631 at t
= 1), bridges 0.49  [gamma = 0.7, uniform start]
2.4   PASS  penalty leakage, projected mixer 0, commutator norms, Strang errors, HPPH contact  [t = 1]
2.5   PASS  a unitary sending six orthogonal states to one output keeps them orthogonal in the environment  [SWAP on C^6
(x) C^6]
2.6   PASS  U_tau = X_last R, U_tau^n = X^(x)n, spectrum = one set of L-th roots of unity per orbit of length L  [n =
2..8]
3.1   PASS  C is the mirror, JC the reversal, a one-bit flip a 120-degree pivot of the tail; |V_m| counts  [every word, m
<= 14 (pivot m <= 11)]
3.2   PASS  the legal one-bit pivot graph is connected  [m = 2..14]
3.3   PASS  C keeps every HP energy; J and P change 48 of 776  [m = 10, HHPHPPHHPPHP]
4.1   PASS  dihedral about bond i is cis iff b_(i-1) = b_i; congruence classes = dihedral words = {x, Cx}; C(a,b) =
(a+b,-b)  [32,764 words (m <= 14); 5,894 valid chains exact (m <= 12)]
4.2   PASS  face(Cx) = 1 - face(x), so (dihedral word, face) returns x; the centre is the middle dihedral  [every word n
= 2..16]
4.3   PASS  turn flip e(j): dihedrals j, j+1; hexagon at i: i, i+4; tail mirror D_k: k; prefix mirror Q_(k-1): k  [m =
6..64]
5.1   PASS  [H, U_C] = 0 and the C-parity basis splits H into H+ (+) H- (residual < 1e-12), with self-loops and parallel
edges kept  [pivot m = 6, 8; hex, hybrid m = 4, 5, 6, 8]
5.2   PASS  a quotient component is frustrated iff its lift holds x and Cx; balanced => equal H+ and H- spectra  [hex,
hybrid m = 4..14; spectra m = 6, 10]
A1    PASS  the largest sorted-eigenvalue difference between H+ and H- is 0.328473 (back2), 0.338886 (both2); the spectra
overlap, so it is not a gap  [m = 10, gamma = 0.7]
```

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

```
5.3   PASS  the alternating route is legal at every width, closes in dihedral words, and ends at the start turned by pi
about the first bond  [heading pattern m <= 257, geometry m <= 64, shape m = 6..20]
5.4   PASS  full and even-sector propagation agree; all-pivot contact 0.0017113, 0.0430001, 0.0335927; hexagon + terminal
start sits in a 2-state island  [m = 10, gamma = 0.7, t in [0, 10]]
6.1   PASS  tail budget k: components 404, 214, 112, 58, 30, 16, 8, 4, 2, 1; the centre cannot change while k < m/2  [m =
10]
6.2   PASS  signatures first collide at 4 turns (0110/1001 only); the rewrite moves beads i+2, i+3, keeps endpoint and
right-turn count  [lengths 1..4; every hexagon edge m <= 12]
6.3   PASS  window 1001/0110 <=> dihedrals trans-cis-trans <=> bonds i and i+4 on one line  [12,304 windows, m = 4..10]
6.4   PASS  codimensions 3, 2, 1, 0/1; 1^m in span iff m = 0 (hex), 0,1 (hybrid), not 3 (back2), always (both2); mirror-
stable components exactly there; legal routes in every class  [span m = 6..257; components m = 4..14; routes m <= 257]
6.5   PASS  bead m never moves under hexagon + terminal (components >= its positions); the m = 10 vocabulary table  [m =
4..14; m = 10 HHPHPPHHPPHP]
A5    PASS  component counts m = 4..14 for both2, back2, hexagon + terminal (against positions of bead m) and hexagon
(against last-bead and turn-count classes)  [m = 4..14]
A2    PASS  every odd-aligned firing flips the pair parities p_j and p_(j+2) only  [m = 10]
7.1   PASS  central parity 0 is necessary, not sufficient; Z_m is one mirror-stable both2 component of 2^((m+4)/2) self-
avoiding, contact-free words (m >= 6); m = 4 fails  [gate m = 10; contacts m = 6..24; component m = 6..20]
7.2   PASS  two mirror-stable both2 components, Z_m and one with phi = 0; Z_m has phi = k mod 2  [m = 6, 10, 14, 18]
A3    PASS  sizes and phi of the two mirror-stable both2 components; all components 2, 14, 106, 858; |V_18| = 125,168  [m
= 6, 10, 14, 18]
7.3   PASS  no 4- or 8-cycles; the 10-cycle catalogue: edges, Z exits, components, masks  [m = 6..14]
7.4   PASS  the small symmetric components at m = 8, 12 are W_m and C(W_m), closed for m = 8..32; 1111||0000 is not a
tau-lock  [m = 8..32]
A6    PASS  the section-148 route from 0000100010 reaches a contact: end, hexagon, end, 10-cycle, end  [m = 10,
HHPHPPHHPPHP]
7.5   PASS  both3 connects V_m for m = 4..21; components and isolated words m = 22..26; residual locks after the 10-
cycle; seven orbits at m = 26  [exhaustive (compiled)]
7.6   PASS  the 22-turn clasp: isolated under both3; the i = 0 exchange frees it; on its reversal the exchange sits at i
= 18; e(3) fills the same four sites  [m = 22]
7.7   PASS  12- and 14-cycle exchanges: m = 6, 8, 10, 12 give 1, 3, 1, 3 then 1, 2, 1, 1; m = 14 stays at 13  [m = 6..14]
8.1   PASS  the half-turn identity; on every serial half-turn the sweep minimum equals the endpoint minimum  [2,000
random cases; 160 half-turns]
8.2   PASS  every hexagon edge is a crankshaft half-turn about the line of its flanking bonds: pure torsion, collision-
free  [697 edges, m = 4..10]
8.3   PASS  the end gate turns the block rigidly about the pivot at angular speed sqrt(2+sigma), lengths and angles
fixed; the in-plane pivot passes a straight angle  [68 gates x 2 senses]
8.4   PASS  two-mirror serial routes: 56 of 68 gates and every lock; 12 gates blocked at both corners; m = 26: 28 of 36
gates, 44 orders  [m = 23..26]
8.5   PASS  certified synchronous clearance >= 0.977 on every gate (exact constant); the planar-radius constant fails on
8 of 72  [68 gates, 360 samples]
8.6   PASS  certified clear in-plane arc: 4/4 (m = 23), 0/4 (24), 4/10 (25), 0/28 (26); every uncertified arc collides
[the 46 locks]
A4    PASS  the three section-150 clasps open in the plane through e(3): the 120-degree pivot of beads 0..3 is certified
clear  [m = 22, 23, 26]
8.7   PASS  R_(pi,x) F(x; phi) = F(Cx; phi) and P_z F(x; phi) = F(x; -phi)  [12 gates, random torsions]
9.1   PASS  exp(-iHt) has no amplitude between components (projector block law)  [m = 10 both2, t = 1, 5, 10]
9.2   PASS  relation layer: K = 2: 21, 30, 8, rank 8, beta1 2; K = 3: 35, 60, 24, rank 22, beta1 4; unsigned rank 24
[tree 01,12,13,34,35,56]
T1    PASS  the running example reads as stated  [x* = 01011110101101011101011110]
ALL CHECKS PASS (44 passed)
```

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